Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/26/2026
PI Name:	VanBuren, Robert Ph.D.		
Project Title:	Leveraging the Natural Resilience of Resurrection Plants for Space Bioregenerative Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 Space Biology NNH24ZDA001N-SBR: E.9 Space Biology Research Studies
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA HQ
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
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Resurrection plants such as Oropetium thomaeum possess the extraordinary ability to survive near complete desiccation by entering a reversible, metabolically quiescent state. Upon rehydration, these plants rapidly resume normal cellular activity, even after prolonged periods in a dry, brittle condition. This remarkable survival strategy is enabled by a suite of protective mechanisms that preserve the structural and functional integrity of cellular macromolecules. These systems include molecular chaperones, antioxidants, osmoprotectants, and vitrifying sugars, which together form a stable intracellular environment that minimizes damage. The same mechanisms may also buffer against damage from other extreme stressors, including ionizing radiation and vacuum exposure, making resurrection plants a compelling model for long term biological preservation and stress resilient crop development in space.
This project will investigate whether desiccation induced macromolecular protection systems in Oropetium can confer cross tolerance to space relevant stresses such as galactic cosmic radiation (GCR) and low pressure environments. By

Task Description:	integrating physiology, microscopy, transcriptomics, metabolomics, and DNA methylation profiling, this proposal aims to (1) identify how desiccation prepares cells for survival in space like conditions, (2) uncover regulatory pathways that support recovery and repair following radiation and vacuum exposure, and (3) determine whether desiccation limits radiation induced damage to DNA and other critical macromolecules. Plants will be exposed to simulated GCR and vacuum at NASA's Space Radiation Laboratory, followed by recovery monitoring and molecular analyses. We will measure the impact of these treatments on photosynthetic performance, cellular integrity, gene regulation, and long-term genome stability. This research will generate a comprehensive dataset linking desiccation tolerance mechanisms to radiation and vacuum resilience, uncovering core stress protective processes that maintain macromolecular integrity. Outcomes will include identification of candidate genes, metabolites, and epigenetic features that contribute to survival and recovery under combined space and desiccation stress. These discoveries will inform new strategies to engineer space hardy crops with enhanced resistance to radiation and desiccation, enabling robust plant based life support systems for future long duration missions. The proposed work directly supports NASA's goals in space crop research and the BLiSS (Biological Life Support Systems) campaign by exploring novel, energy efficient methods of preserving biological function in extreme environments. Unlike traditional cryopreservation, resurrection plant xeropreservation relies on biologically encoded stabilization of macromolecules in the dry state, offering a potential alternative for long term tissue storage and transport without ultra low temperature requirements. These mechanisms may be especially valuable for minimizing infrastructure and energy demands in closed loop bioregenerative systems. This foundational research will guide the functional validation of protective compounds and regulatory proteins identified in resurrection plants. Genetic and biochemical tools will be used to determine their roles in mitigating radiation damage and preserving cellular homeostasis. Promising macromolecular protective factors will be engineered into crop species to assess their utility for improving stress resilience in space environments. In parallel, future research will explore the limits of radiation tolerance in resurrection plants and refine strategies for synthetic xeropreservation. These combined efforts will expand the capacity to deploy resilient plant systems in extraterrestrial settings, ultimately supporting sustained human presence beyond Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will uncover molecular and physiological mechanisms that enable resurrection plants to withstand extreme dehydration, ionizing radiation, and vacuum, which are stressors analogous to space environments. We aim to identify natural protective systems that stabilize cellular structures and preserve viability without water or atmosphere. Expected outcomes include a systems-level understanding of stress resilience, key protective metabolites and repair genes, and regulatory networks governing rehydration and recovery. Deliverables will include open-access multi-omics datasets, microscopy data, and models linking physiology to molecular responses. These insights will support biopreservation technologies, long-term tissue storage, and development of space-resilient crops for life support in space.
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated:)